

REAL-TIME EMPLOYEE PRESENCE AND PREDICTIVE ANALYTICS SOLUTION

A Project Report

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by

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1 INTRODUCTION

1.1 Project Origin

The genesis of this project is deeply rooted in the high-stakes, zero-slack environment of a medium-sized packaging manufacturing facility where labor is not merely a cost, but the primary engine of operational throughput. During an intensive internship observation, it became evident that while the facility's machinery was meticulously maintained and calibrated, the "human component" was managed through archaic, non-integrated tools that failed to account for the volatility of workforce availability. The organization operates on relentless, high-volume production schedules that demand perfectly synchronized shift work; consequently, when raw materials are staged and high-speed printing or cutting machines are calibrated based on the rigid assumption of full staff capacity, even a minor 5% unexpected absence rate can trigger a disproportionate 20% drop in overall line efficiency.

1.1.1 Major Areas of Current Focus

The primary heartbeat of the firm's operational management is the maintenance of a continuous, uninterrupted production flow, which requires a hyper-focus on minimizing machine downtime and optimizing the consumption of raw materials. Currently, shift supervisors and floor managers spend the critical first hour of every rotation in a state of reactive "firefighting," manually verifying the presence of specialized engineering personnel and reassigning tasks on the fly to prevent catastrophic supply chain bottlenecks. This focus on immediate crisis management creates a "Butterfly Effect" where a single missing operator in an upstream department like printing causes idle time in downstream departments like folding and gluing, ultimately leading to late delivery penalties and reduced machine life due to emergency stops and restarts.

1.1.2 Overview of the Project

To mitigate these systemic vulnerabilities, this project proposes the development of a specialized software application that functions as a centralized intelligence hub and a "Single Source of Truth" for the entire organization. By implementing role-based dashboards tailored specifically for Super Admins, Sub-Admins, and Employees, the platform replaces fragmented, disjointed manual tracking with an automated, proactive system that digitizes the entire labor lifecycle. Beyond simple time-tracking, the system introduces a revolutionary predictive layer that forecasts future workforce availability, effectively moving the "awareness window" from ten minutes after a shift starts to twenty-four hours before it begins, thereby allowing management to secure the necessary engineering talent well in advance of production.

2 SYSTEM STUDY

2.1 Competency Center Overview

A thorough study of existing HR and Operations hubs within medium-sized manufacturing reveals that these "competency centers" are often misnomers, acting more as data graveyards where information is recorded for clerical reasons but never utilized for strategic planning. These centers rely heavily on manual data processing, physical muster rolls, and disconnected spreadsheet dumps that, while technically sufficient for retroactive payroll calculation, completely lack the integrated analytical tools necessary for real-time operational navigation. This information asymmetry means that a production manager often remains unaware that an employee was granted leave by the HR office until the machinery is already powered on, leading to significant wasted "Green-to-Green" time—the period when machines are actually generating value.

2.2 Need for the System

The proposed system is designed to bridge the costly "Production Gap," which represents the lost productivity occurring when a shift is understaffed because absences were recorded only after the work day commenced. Existing market solutions like SAP or Oracle are often prohibitively expensive and "over-engineered" for small to medium enterprises, requiring months of specialized training and massive licensing fees without offering the specific predictive granularity needed for shop-floor logistics. This project fills a vital niche by providing a lightweight yet powerful tool that anticipates individual attendance patterns, giving managers the lead time required to call in cross-trained staff or adjust machine schedules before a shortage can paralyze the production line.

2.3 Outline of the System

The system resolves these operational gaps through a sophisticated, four-pillar analytical journey that moves the organization from reactive observation to predictive intelligence. It begins with descriptive analytics via real-time dashboards for the immediate visualization of current presence, followed by diagnostic tracking of historical leave patterns to identify root causes, such as absenteeism spikes on specific night shifts or following public holidays. Building on this foundation, the predictive pillar utilizes self-developed algorithms to calculate "Presence Probability Scores," while the prescriptive pillar provides managers with actionable, data-driven insights required to proactively reallocate shifts and balance resources, ensuring the production floor remains resilient against human volatility.

3 SYSTEM ANALYSIS

3.1 System Objectives

The primary objectives of this system are to securely automate the daily employee clock-in and clock-out processes while totally digitizing the leave application and approval workflow to eliminate administrative friction and paper waste. The solution is specifically engineered to replace "guesswork with framework" by accurately forecasting individual employee presence for future shifts based on weighted historical behavior. Furthermore, the system aims to enhance corporate transparency and accountability by generating dynamic, multi-page PDF reports that summarize workforce statistics, enabling high-level management to conduct comprehensive reviews and make informed, data-backed decisions regarding labor capacity and shift scaling.

3.2 Scope of the System

The scope of the system is intentionally focused on internal workforce management for small to medium-sized enterprises (SMEs) to ensure maximum performance and ease of deployment without the "feature bloat" of enterprise resource planning suites. It covers the entire operational lifecycle, including user onboarding, granular shift assignment, daily time tracking, and leave balance deductions. To maintain a lean architectural profile and avoid the regulatory complexities of financial software, the system does not cover external recruitment or payroll processing, focusing exclusively on the "Operational Uptime" of the current workforce.

3.3 Functional Requirements

3.3.1 Description of Inputs and Outputs

The system manages a complex array of inputs, including encrypted user credentials, biometric or manual timestamps, leave application parameters, and supporting digital documents like medical certificates in image or PDF format. These inputs are processed to generate highly functional outputs, such as validated session tokens for secure access, interactive UI charts including Doughnut and Bar graphs for visual heatmaps of attendance trends, and real-time approval status updates. Most critically, the system generates a "Presence Probability Score" (Pp) ranging from 0-100% for each employee, which is presented alongside downloadable, professional-grade statistical reports that serve as the formal documentation for shift planning meetings.

3.3.2 System Functionality

The core functionality is governed by a strict Role-Based Access Control (RBAC) model that ensures data security and operational clarity across the organization. The Super Admin is granted global control over the system architecture, audit logs, and high-level user management; the Sub-Admin, or floor manager, handles team-specific leave approvals and utilizes the predictive modeling tools for upcoming shift planning. Employees interact with a self-service portal designed to simplify the logging of daily time and the submission of leave requests, providing them with a clear view of their own attendance metrics and fostering a culture of professional accountability.

3.4 Design Constraints

The system must operate under several critical design constraints to remain viable for the target manufacturing sector, particularly the need for extreme cost-effectiveness by utilizing open-source frameworks like Python and Flask rather than expensive proprietary software. Operationally, the system is designed for high-speed execution, requiring that prediction logic and data retrieval occur within seconds to prevent "spinner fatigue" for busy managers. From a

data integrity standpoint, the architecture strictly enforces a "one record per user per day" rule for attendance, which prevents the data corruption and inflated labor statistics often associated with manual systems or poorly designed digital logs.

3.5 Proposed System

Formally titled the "Real-Time Employee Presence and Predictive Analytics Solution," this system stands as a specialized, manufacturing-centric alternative to generalized HR applications like BambooHR or Zoho People. Unlike those platforms, which primarily serve as digital filing cabinets for historical employee records, this system prioritizes the proactive forecasting of labor as a critical production variable. By transforming raw attendance data into actionable predictive insights, it empowers management to move from a defensive posture to a strategic one, anticipating absenteeism and reallocating human capital to prevent costly production stalls before a shift even begins.

3.6 Environmental Model

The system is built upon a modern, three-tier architecture that separates the presentation layer from the core logic and data storage tiers for enhanced scalability. The server environment is powered by Python 3.9+ and the Flask micro-framework, utilizing a relational database model that begins with SQLite for development and is fully compatible with PostgreSQL for enterprise-level scaling. The client environment is accessible via any modern, JavaScript-enabled web browser to support the interactive UI, while the data science environment leverages the power of Pandas, NumPy, and Statsmodels for executing the self-developed predictive algorithms and Matplotlib for rendering visual statistical data.

3.7 Behavioral Model

The system functions as a robust state machine, transitioning through clearly defined states based on specific user interactions and data triggers. It initiates in a validation state whenever an employee interacts with the clock-in module to ensure data integrity and security, before moving into a data-persistence state to permanently update the primary attendance records. For forecasting tasks, the system enters a heavy-lifting processing state where it retrieves historical logs, applies decay-weighted algorithms to account for recent behavior, and finally reaches a presentation state to render the "Presence Probability Scores" and generate the necessary management reports.

3.8 Conceptual Data Model

The conceptual data model is meticulously structured around four primary entities—User, Shift, Attendance, and LeaveRequest—which are normalized to eliminate data redundancy and ensure relational integrity. The User entity stores hashed credentials and role assignments, while the Shift entity defines the operational timing boundaries; the Attendance entity logs real-time presence with unique composite key constraints to prevent duplication. The LeaveRequest entity manages the digital lifecycle of planned absences, providing the "future-dated" data points that, when combined with historical attendance, allow the self-developed algorithms to generate the high-fidelity predictive insights that define the system's value.

3.9 Workflow

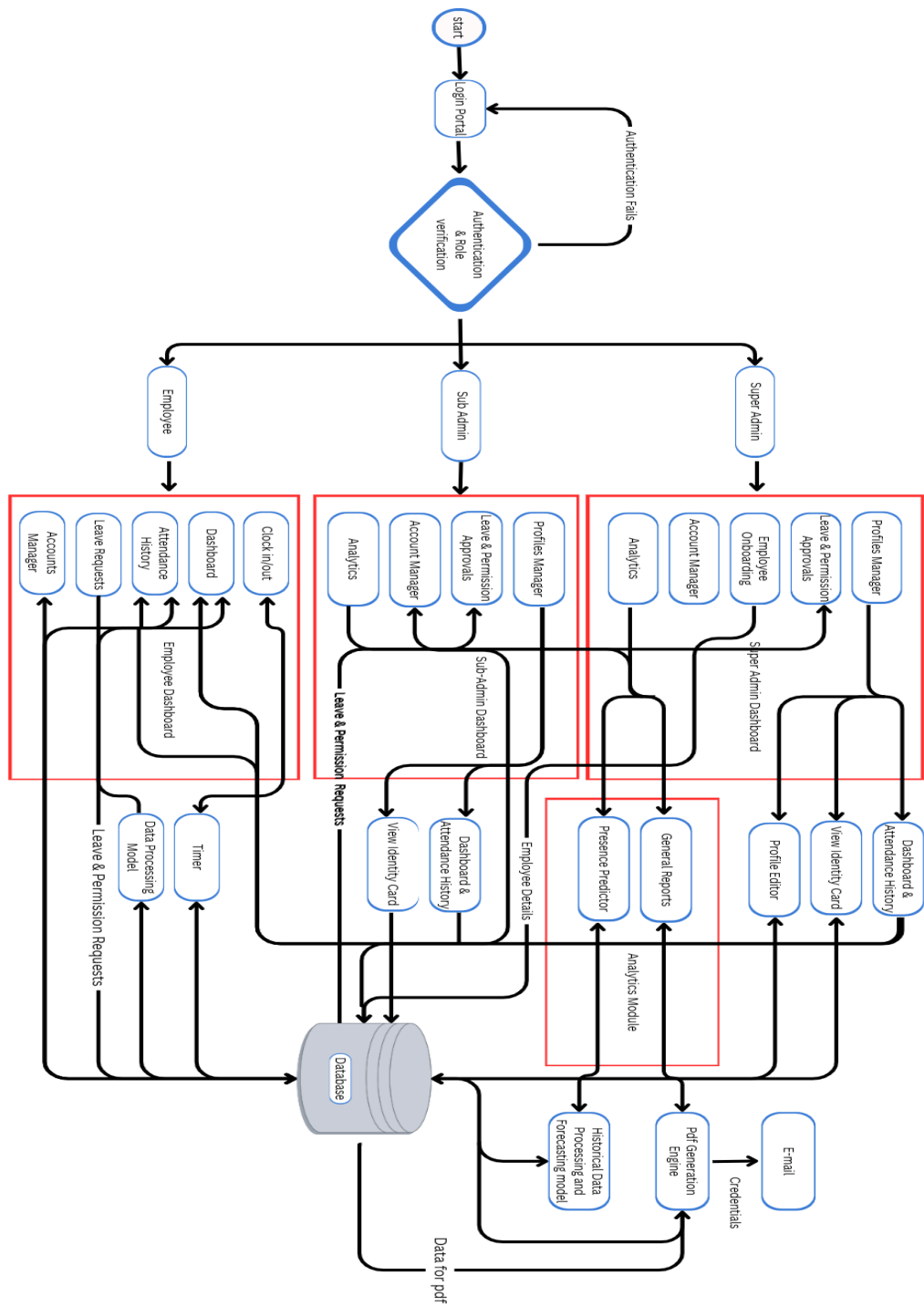


Figure 3.1 Project Workflow

4. SYSTEM DESIGN

4.1 Design Methodology

The system utilizes the Model-View-Controller (MVC) pattern to ensure a clean separation between data logic and the user interface. Flask serves as the Controller by managing routing and application flow, while SQLAlchemy acts as the Model to handle the database schema and maintain data integrity for the predictive algorithms. The View is powered by Jinja2 and Bootstrap 5.3, which work together to render a responsive, mobile-friendly frontend that transforms complex workforce analytics into intuitive visual dashboards for real-time management.

4.2 Table Design

Attribute	Type	Description
id	Primary Key	Unique identifier for each employee.
name	String	The full name of the employee.
email	String (Unique)	Corporate email used for login and notifications.
password	String (Hashed)	Securely stored password using SHA-256 hashing.
role	String	Defines access level (Super Admin, Sub-Admin, Employee).
team	String	The specific department or production line assigned.
cl_balance	Integer	Current Casual Leave quota available.
ml_balance	Integer	Current Medical Leave quota available.
shift_id	Foreign Key	Links the user to their standard working hours in the Shift table.

Table 4.1 User Table

Attribute	Type	Description
id	Primary Key	Unique identifier for the shift type.
name	String	Label for the shift (e.g., Morning, Evening, Night).
start_time	Time	The official start time for attendance validation.
end_time	Time	The official end time for the workday.

Table 4.2 Shift Table

Attribute	Type	Description
id	Primary Key	Unique identifier for the attendance record.
user_id	Foreign Key	Reference to the specific employee.
date	Date	The calendar day of the record.
in_time	DateTime	The specific timestamp of the "Clock In" event.
out_time	DateTime	The specific timestamp of the "Clock Out" event.
status	String	Categorization of presence (Present, Late, Absent).
shift_id	Foreign Key	The shift the employee was expected to work.

Table 4.3 Attendance Table

Attribute	Type	Description
id	Primary Key	Unique identifier for the request.
user_id	Foreign Key	Reference to the employee applying for leave.
type	String	Category of leave: Casual (CL), Medical (ML), or On-Duty (OD).
status	String	Current state: Pending, Approved, or Rejected.
attachment	String/Blob	Reference to uploaded evidence, such as medical certificates.

Table 4.4 Leave Request Table

4.3 Entity Relationship Diagram

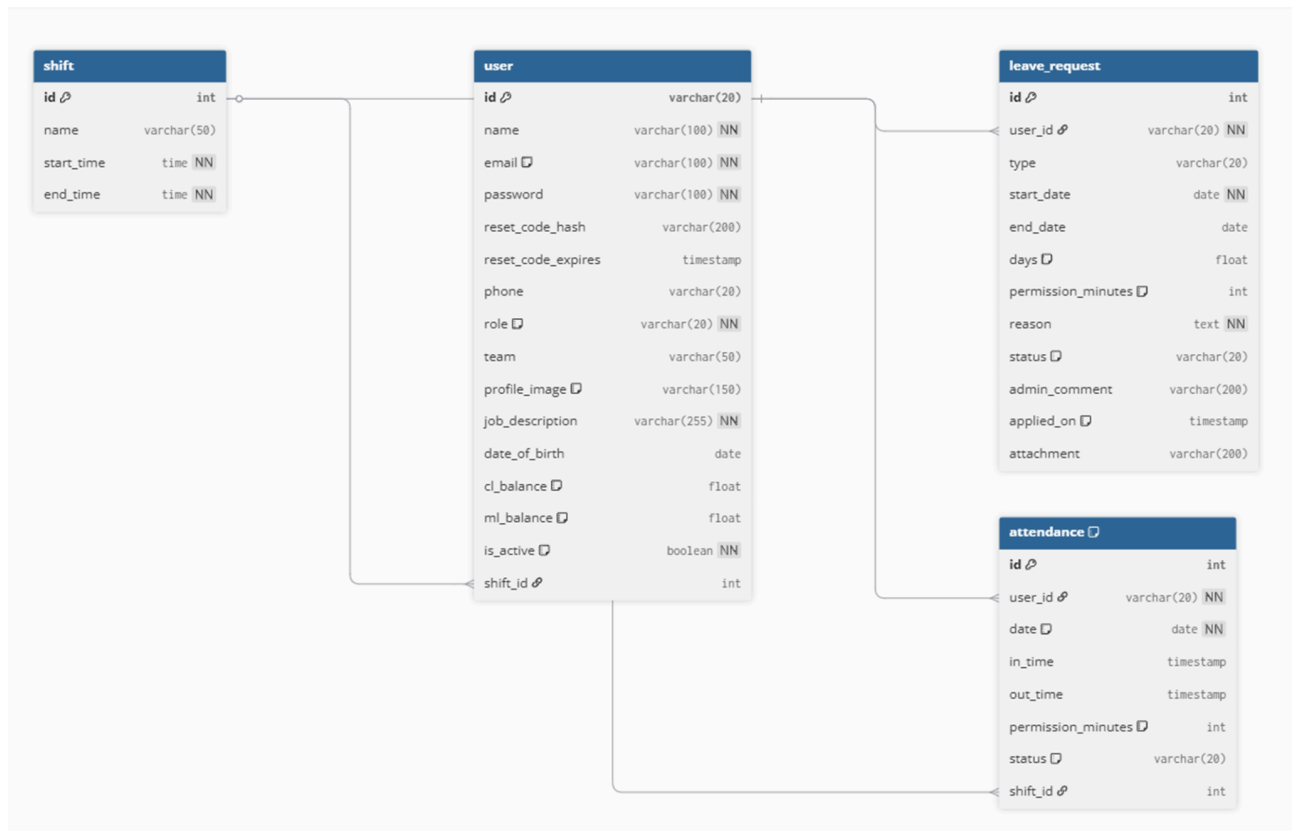


Figure 4.1 Entity Relationship Diagram

4.3.1 Shift and Attendance Tables

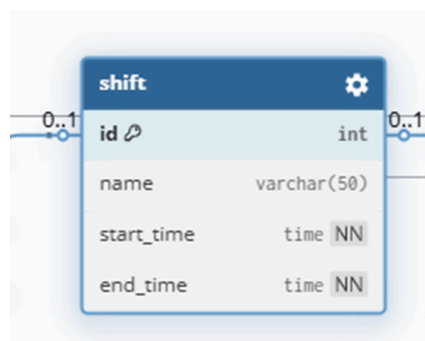


Figure 4.2 Shift Table

4.3.2 User Table

user	
id 🔗	varchar(20)
name	varchar(100) NN
email 📧	varchar(100) NN
password	varchar(100) NN
reset_code_hash	varchar(200)
reset_code_expires	timestamp
phone	varchar(20)
role 📋	varchar(20) NN
team	varchar(50)
profile_image 📷	varchar(150)
job_description	varchar(255) NN
date_of_birth	date
cl_balance 📊	float
ml_balance 📊	float
is_active 📋	boolean NN
shift_id 🔗	int

Figure 4.3 User Table

4.3.1 Attendance Table

attendance		
id		int
user_id	varchar(20)	NN
date	date	NN
in_time	timestamp	
out_time	timestamp	
permission_minutes		int
status	varchar(20)	
shift_id	int	

Figure 4.4 Attendance Table

4.3.1 Leave Request Table

leave_request		
id		int
user_id	varchar(20)	NN
type	varchar(20)	
start_date	date	NN
end_date	date	
days		float
permission_minutes		int
reason	text	NN
status	varchar(20)	
admin_comment	varchar(200)	
applied_on	timestamp	
attachment	varchar(200)	

Figure 4.5 Leave Request Table

4.4 Output Design

The system's output architecture is meticulously engineered for immediate operational readability, ensuring that floor managers can interpret complex workforce data at a single glance to make high-stakes staffing decisions. Individual presence is distilled into "Probability Scores" presented in a clear, prioritized list format, allowing supervisors to identify high-risk attendance gaps before a shift commences. For deeper organizational oversight, the system leverages the **FPDF** library to programmatically compile structured, multi-page PDF documents that serve as formal analytical reports. These documents are enriched with embedded **Matplotlib** charts, which translate raw historical attendance and leave data into intuitive visual trends, providing a professional-grade synthesis of workforce metrics that can be easily shared or archived for long-term strategic review.

5 MODULE DESIGN

5.1 Module Specification

The system architecture is anchored in a modular framework designed to prioritize scalability, security, and a clean separation of concerns across the organizational hierarchy. By partitioning the application into three primary functional areas—specifically mapped to the Role-Based Access Control (RBAC) matrix for Employees, Sub-Admins, and Super Admins—the design ensures that data processing remains isolated and highly secure. This structural modularity not only optimizes the efficiency of UI component loading but also provides a "future-proof" foundation, allowing developers to implement advanced feature enhancements or integrate new predictive models without the risk of disrupting core system stability or crossing permission boundaries.

5.2 Module Description

5.2.1 Super Admin Module

Positioned at the apex of the access hierarchy, the Super Admin Module is engineered for high-level strategic planning and comprehensive organizational oversight. Its Automated Onboarding utility revolutionizes the induction process by instantly generating secure, downloadable PDF credentials that include initial login details and shift parameters, significantly reducing the administrative burden on HR departments during new employee transitions. Beyond onboarding, this module serves as the primary engine for executive intelligence, featuring a Dynamic Reporting component that synthesizes vast streams of predictive forecasts and historical logs into professional, multi-page PDF documents on-the-fly, giving top-tier management a "God's-eye view" of workforce trends and system-wide health..

5.2.2 Sub-Admin Module

The Sub-Admin Module acts as the operational nerve center for mid-level management, providing the granular tools necessary for real-time team supervision and proactive shift planning. Through a centralized Approvals Block, supervisors can meticulously review pending leave requests and verify digital medical certificates with a single click, triggering an automated update of leave balances within the central database to maintain a strict single source of truth. The module's most innovative feature is the Forecasting Engine, which allows managers to input future dates and execute self-developed algorithms to generate Presence Probability Scores (0% to 100%); this serves as a vital early warning system, enabling managers to identify potential labor shortages and reallocate resources before a production line stall can occur.

5.2.3 Employee Module

Designed as a secure and intuitive self-service portal, the Employee Module is the primary touchpoint for the general workforce to manage their daily attendance and leave lifecycle. The core Attendance Engine facilitates a streamlined "Clock In/Out" process that enforces a strict "single punch per day" constraint, which is fundamental to maintaining the data integrity required for accurate downstream predictive analytics. To foster transparency and engagement, the module features personal Visual Dashboards—highlighted by real-time Doughnut charts—that provide employees with an immediate, clear breakdown of their monthly attendance statistics and remaining leave quotas, while a secure digital form simplifies the process of applying for time off and uploading supporting documentation.

6 TESTING

6.1 Unit Testing

Unit testing serves as the fundamental layer of the quality assurance process, focusing on the rigorous verification of individual logic blocks in total isolation to ensure they perform exactly as intended before they are integrated into the broader system. This phase is critical for validating high-stakes security functions, such as the Werkzeug library's ability to correctly hash and salt passwords during the user creation process, which prevents plain-text credentials from ever touching the database. Additionally, it confirms the structural integrity of the **UniqueConstraint** within the Attendance table, verifying that the system successfully rejects any attempts to record multiple clock-in events for a single user on the same date, thereby safeguarding the accuracy of the datasets that will eventually feed the predictive algorithms.

6.2 Integration Testing

Integration testing evaluates the "handshakes" and data dependencies between disparate modules, ensuring that the system functions as a cohesive unit rather than a collection of siloed parts. The primary objective is to verify the accuracy of automated data propagation; for example, the test confirms that when a Sub-Admin approves a leave request within their dedicated module, the corresponding `cl_balance` or `ml_balance` (casual or medical leave) in the User table is decremented instantly without manual intervention. This level of testing also ensures that these dynamic database updates are correctly indexed and retrieved by the reporting engine, resulting in a FPDF-generated Analytics Report that perfectly reflects the most current leave statuses and attendance records across the organization.

6.3 System Testing

System testing acts as the final, comprehensive stress test of the application, simulating an exhaustive end-to-end workflow to ensure the software can withstand the pressures of a live manufacturing environment. This phase mimics a complete operational cycle: an Admin initializes a new employee profile, the employee logs multiple days of attendance and submits a leave request, and the manager navigates the approval process. The test culminates in a high-load query to the Predictive Module for a future date, verifying that the self-developed algorithms can calculate and return a "Presence Probability Score" within a strictly defined latency window. This ensures the entire architecture is resilient, responsive, and free from server timeouts, providing a reliable tool for real-time shift planning and production floor management.

7 CONCLUSION

By replacing manual, reactive logging with a unified and intelligent platform, this project successfully bridges the critical "Production Gap" that has historically plagued medium-sized manufacturing operations. This software empowers shift managers to transcend the limitations of retroactive data, allowing them to move the "awareness window" from ten minutes after a shift starts to twenty-four hours before it begins, thereby foreseeing potential staffing shortages before they manifest as idle machinery or downstream bottlenecks. By enabling proactive resource reallocation, the system ensures that high-volume production schedules remain perfectly synchronized, directly contributing to the conservation of expensive raw materials and the prevention of costly downtime. Ultimately, this transformation from a defensive management posture to a data-driven offensive strategy turns workforce presence into a predictable asset rather than a volatile liability.

7.1 Future Enhancements

Looking ahead, the system is designed with an extensible architecture that paves the way for seamless integration with physical biometric hardware, such as RFID or fingerprint scanners, to transition the "Clock-In" process from a web portal into the physical realm of the factory floor. This hardware-level automation would eliminate manual entry entirely, further reducing the risk of administrative errors while increasing the velocity of shift handovers during high-stakes transitions. Additionally, the implementation of an automated notification layer—utilizing SMS or email triggers—will allow the system to act as an autonomous sentinel, instantly alerting managers the moment a key employee's "Presence Probability Score" (SP_p) falls below a pre-defined safety threshold. This ensures that human intervention occurs precisely when and where it is most needed, maintaining operational resilience without requiring constant manual monitoring.

7.2 Scope of the Project

While the initial requirements were meticulously tailored to the unique operational pressures of a packaging manufacturing firm, the core predictive engine and robust leave management framework possess a high degree of industrial generalizability. The software's underlying logic is built to handle the complexities of any environment reliant on strict, time-sensitive shift schedules, making it a powerful solution for sectors far beyond the factory floor. This broad scope suggests successful implementation opportunities in healthcare for nursing rotations, logistics for high-volume warehouse dispatching, and large-scale retail operations, where the ability to anticipate workforce availability is just as critical to maintaining service levels and operational continuity as it is in a production environment

APPENDIX 1: System Screenshots

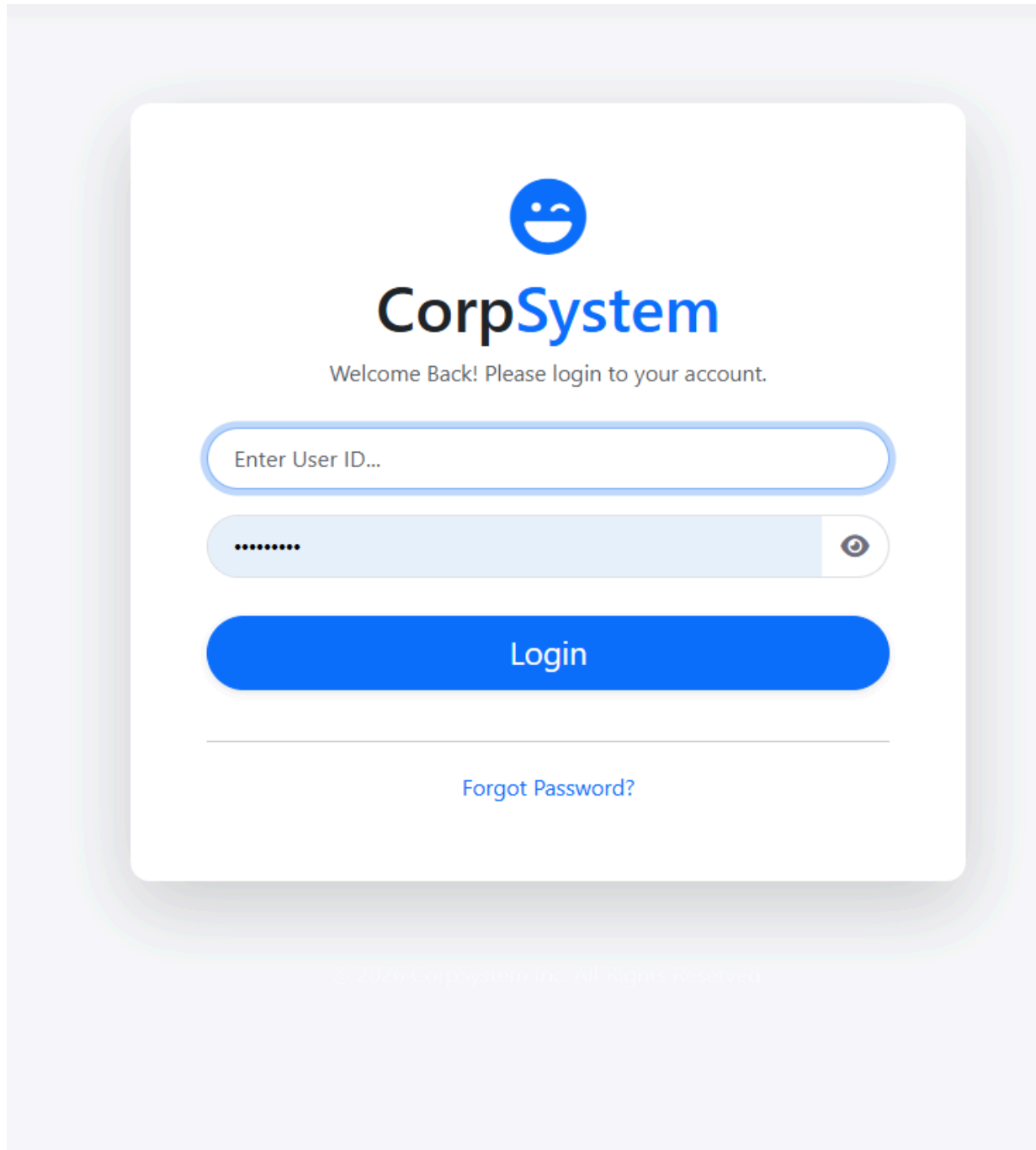


Fig. A-1.1 Login Page

 CorpSystem

Super Admin

Analytics

Logout

Super Admin Console

Profiles

Approvals

Add Employee

Settings

Employee Directory

Search for name...

USER	ROLE	TEAM	STATUS	ACTIONS
SA Super Admin ADM001	SuperAdmin	Management	Active	
JD John Doe EMP001	Employee	Team 1	Active	
JS Jane Smith EMP002	Employee	Team 1	Active	
AB Alice Brown EMP003	Employee	Team 2	Active	
BW Bob White EMP004	Employee	Team 2	Active	
SOMU SUNDARAM ADM002	SubAdmin	Team 2	Active	

Fig. A-1.2 Super Admin. Module - Profiles Manager

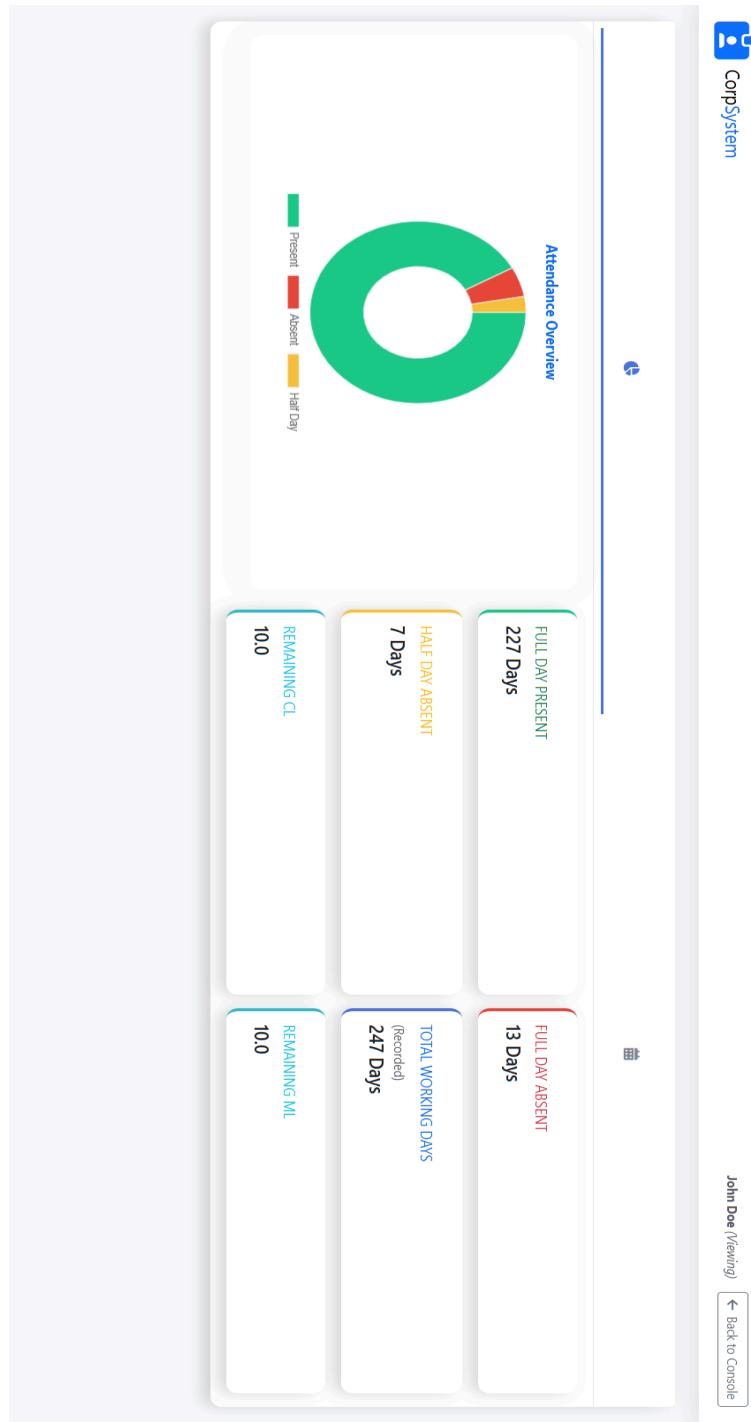


Fig. A-1.3 Super Admin. Module - Employee Statistics

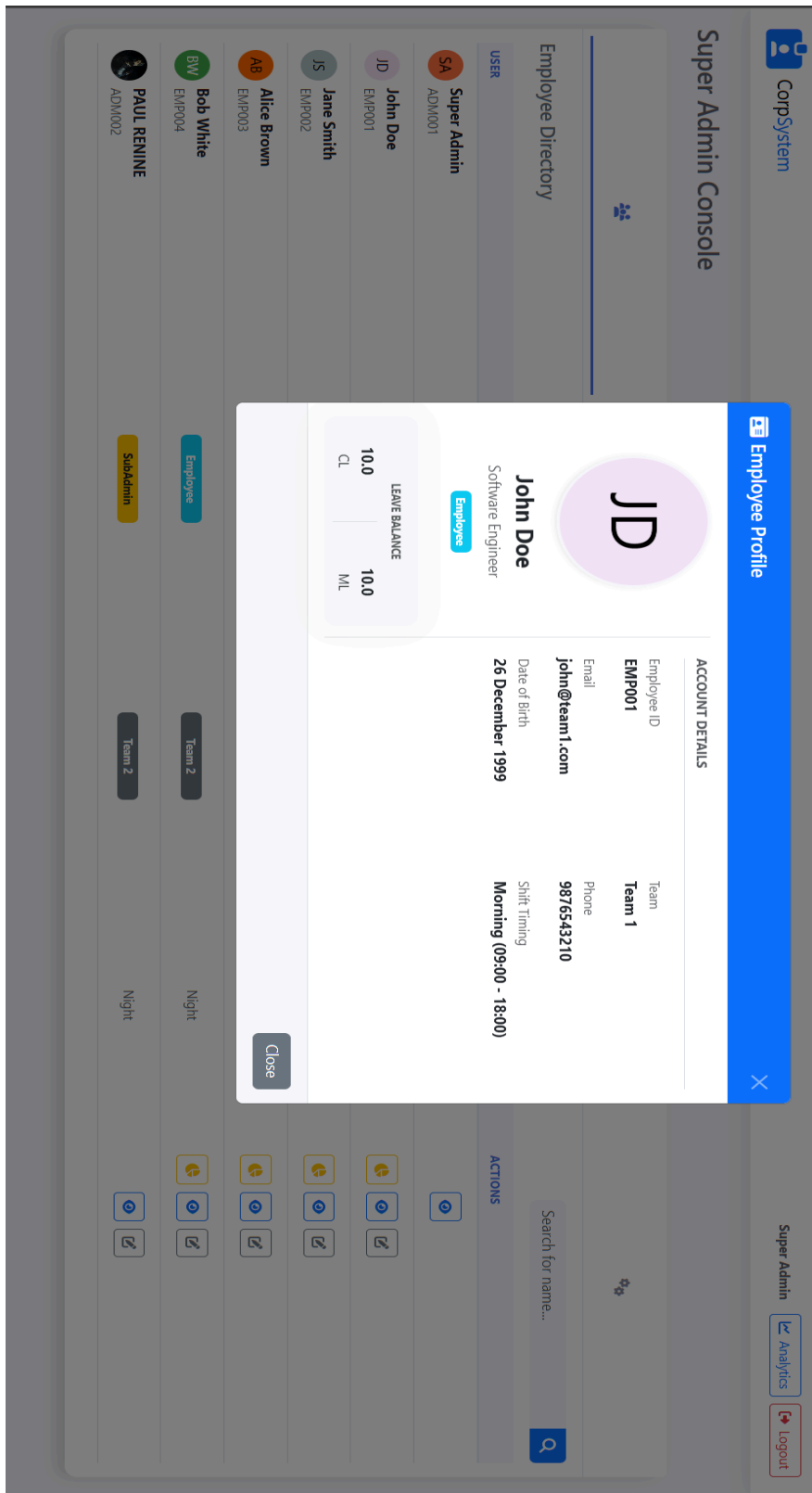


Fig. A-1.4 Super Admin. Module - Profiles Details

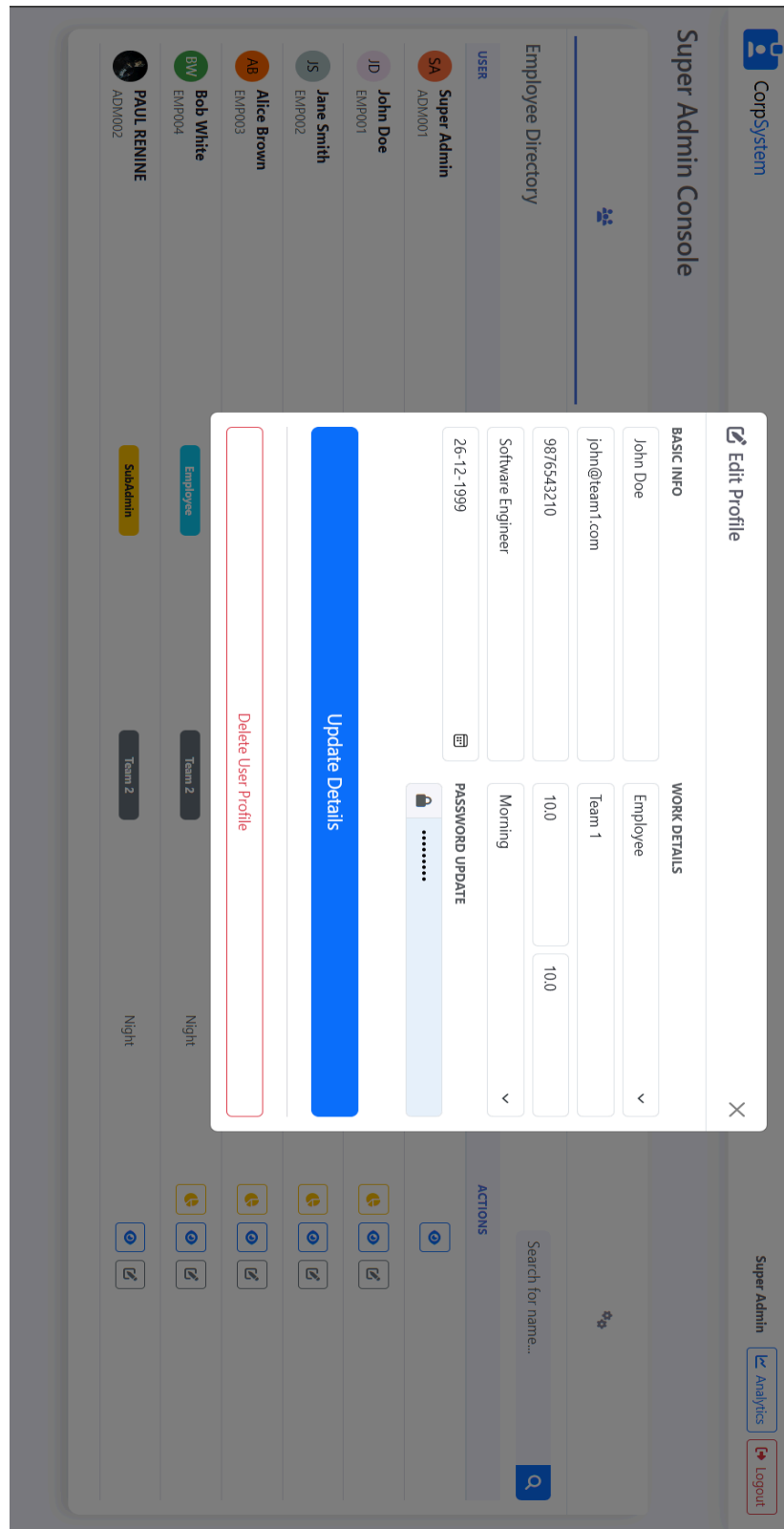


Fig. A-1.5 Super Admin. Module - Profiles Editor



CorpSystem

Super Admin

← Back to Console

Logout

Analytics & Reporting

General Reports

Presence Predictor

USER	ROLE	TEAM	REPORT ACTION
Super Admin ADM001	SuperAdmin	Management	Generate Report
John Doe EMP001	Employee	Team 1	Generate Report
Jane Smith EMP002	Employee	Team 1	Generate Report
Alice Brown EMP003	Employee	Team 2	Generate Report
Bob White EMP004	Employee	Team 2	Generate Report
PAUL RENINE ADM002	SubAdmin	Team 2	Generate Report

Fig. A-1.7 Super Admin. Module - Analytics Page 1

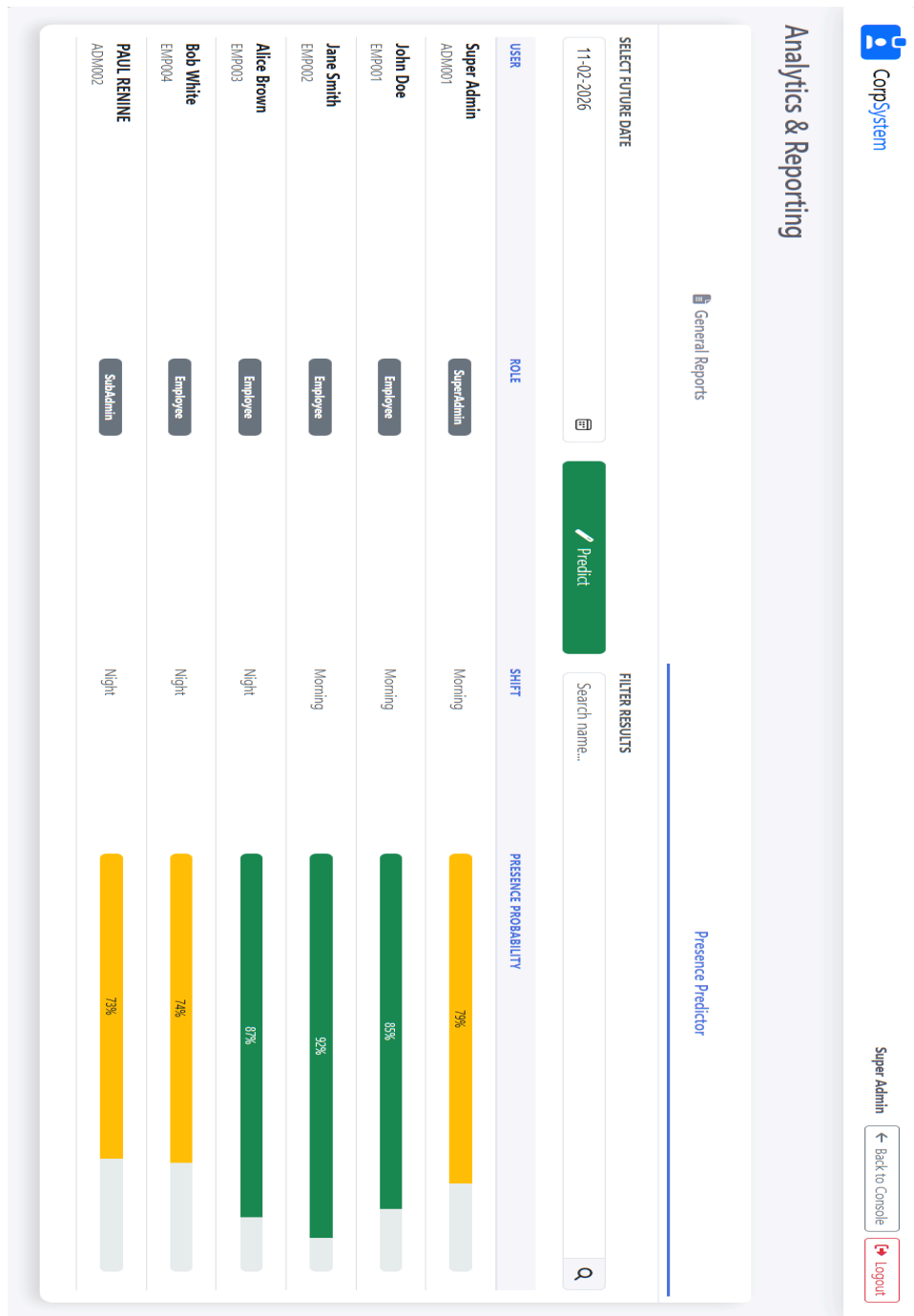


Fig. A-1.8 Super Admin. Module - Analytics Page 2

Corpsystem

PAUL RENINE

Analytics

Logout

Team Manager Console

Team Directory

USER

ROLE

TEAM

SHIFT

ACTIONS

Alice Brown
EMP003

Employee

Team 2

Night



Bob White
EMP004

Employee

Team 2

Night



PAUL RENINE
ADM002

SubAdmin

Team 2

Night



Search team members...



Fig. A-1.9 Sub Admin. Module - Profiles Manager

30

 Corpsystem

PAUL RENINE

[← Back to Console](#)[Logout](#)

Analytics & Reporting

General Reports

Presence Predictor

USER	ROLE	TEAM	REPORT ACTION
Alice Brown EMP003	Employee	Team 2	Generate Report
Bob White EMP004	Employee	Team 2	Generate Report
PAUL RENINE ADM002	SubAdmin	Team 2	Generate Report

Fig. A-1.10 Sub Admin. Module - Analytics Page 1

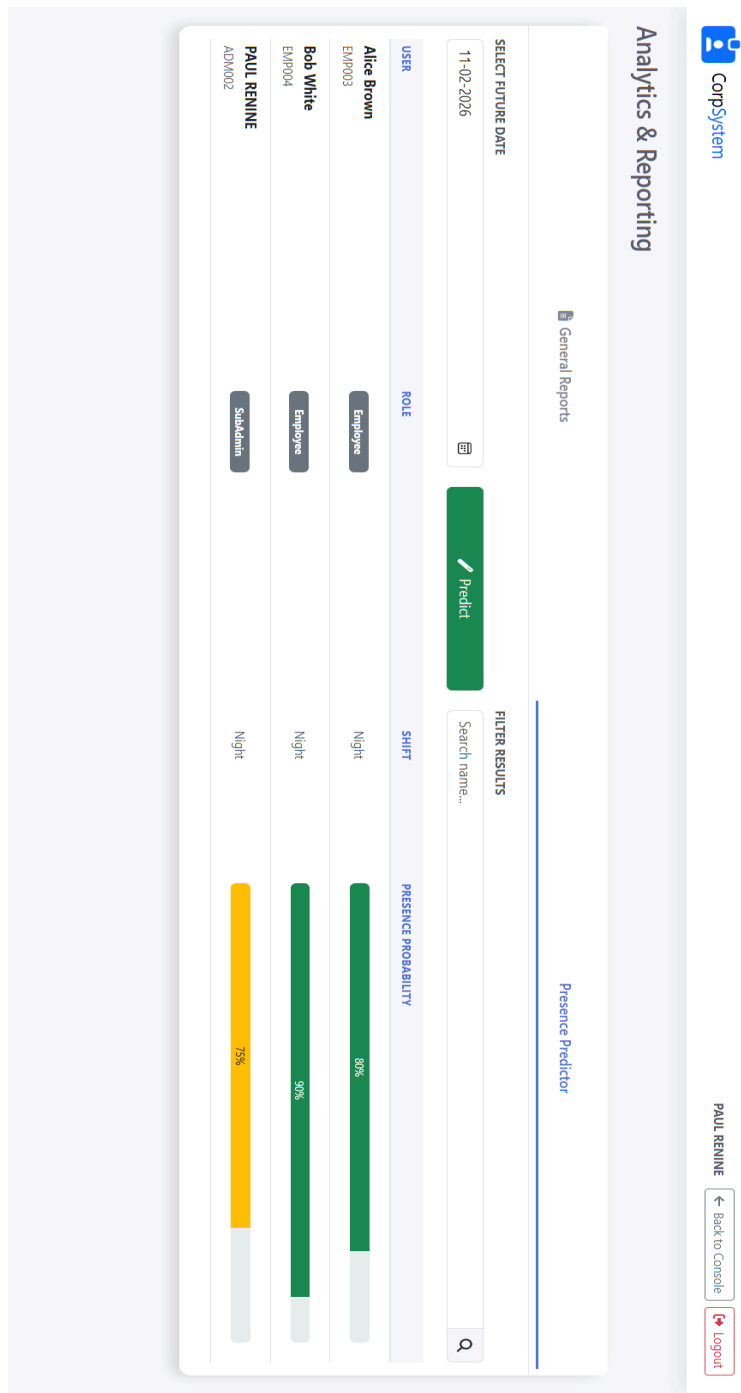


Fig. A-1.5 Sub Admin. Module - Analytics Page 2

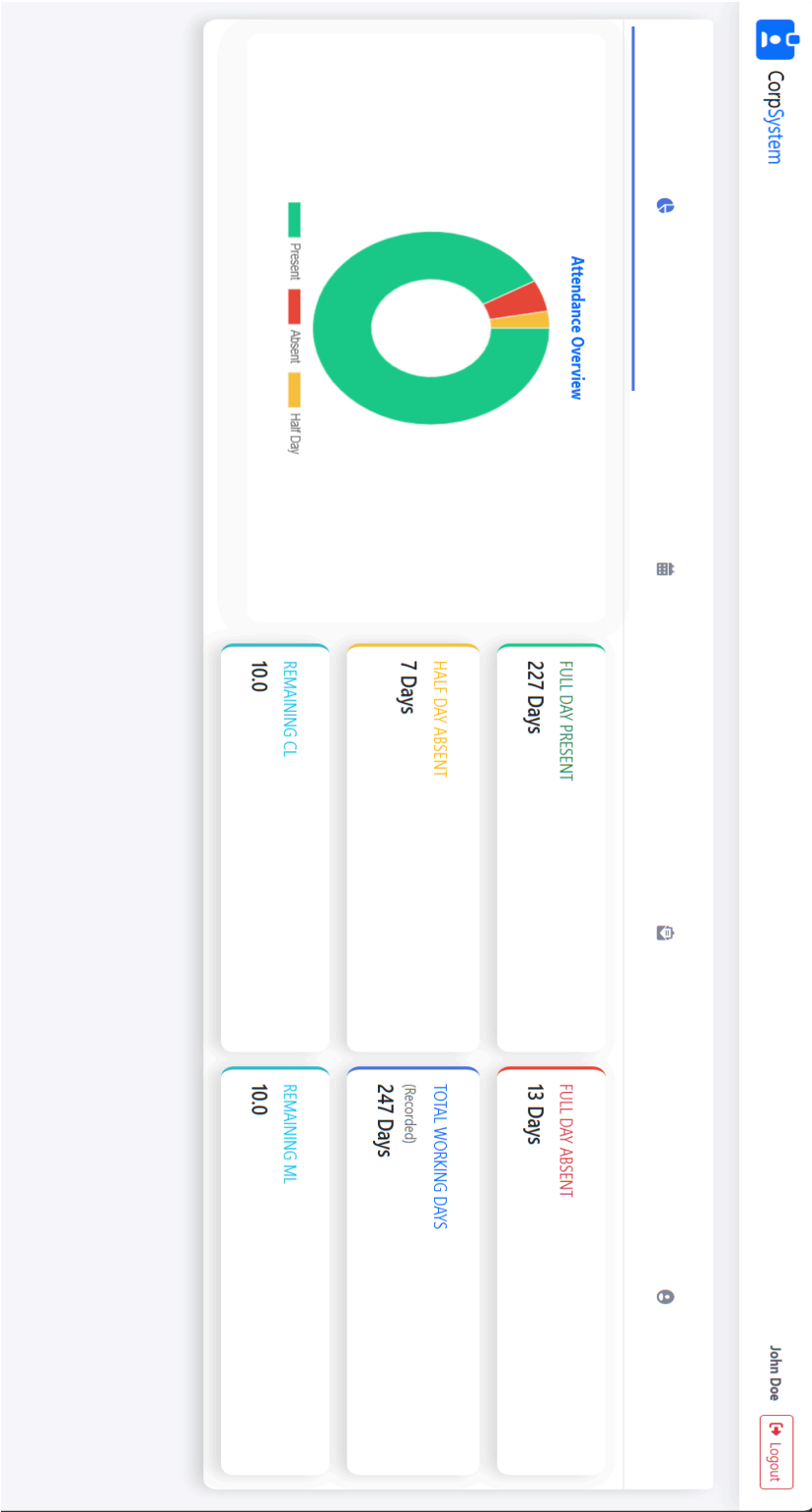


Fig. A-1.11 Employee Module - Dashboard

 Corpsystem

John Doe

Logout

DATE	DAY	SHIFT DETAILS	IN TIME	OUT TIME	STATUS
2026-01-20	Tuesday	Morning 09:00 - 18:00	09:00	18:13	Present
2026-01-19	Monday	Morning 09:00 - 18:00	09:14	13:00	Half Day
2026-01-16	Friday	Morning 09:00 - 18:00	09:15	13:00	Half Day
2026-01-15	Thursday	Morning 09:00 - 18:00	09:12	18:27	Present
2026-01-14	Wednesday	Morning 09:00 - 18:00	--:--	--:--	Absent
2026-01-13	Tuesday	Morning 09:00 - 18:00	09:01	18:17	Present
2026-01-12	Monday	Morning 09:00 - 18:00	09:10	18:26	Present
2026-01-09	Friday	Morning 09:00 - 18:00	09:04	18:07	Present
2026-01-08	Thursday	Morning 09:00 - 18:00	09:07	18:28	Present

Fig. A-1.12 Employee Module - Attendance History

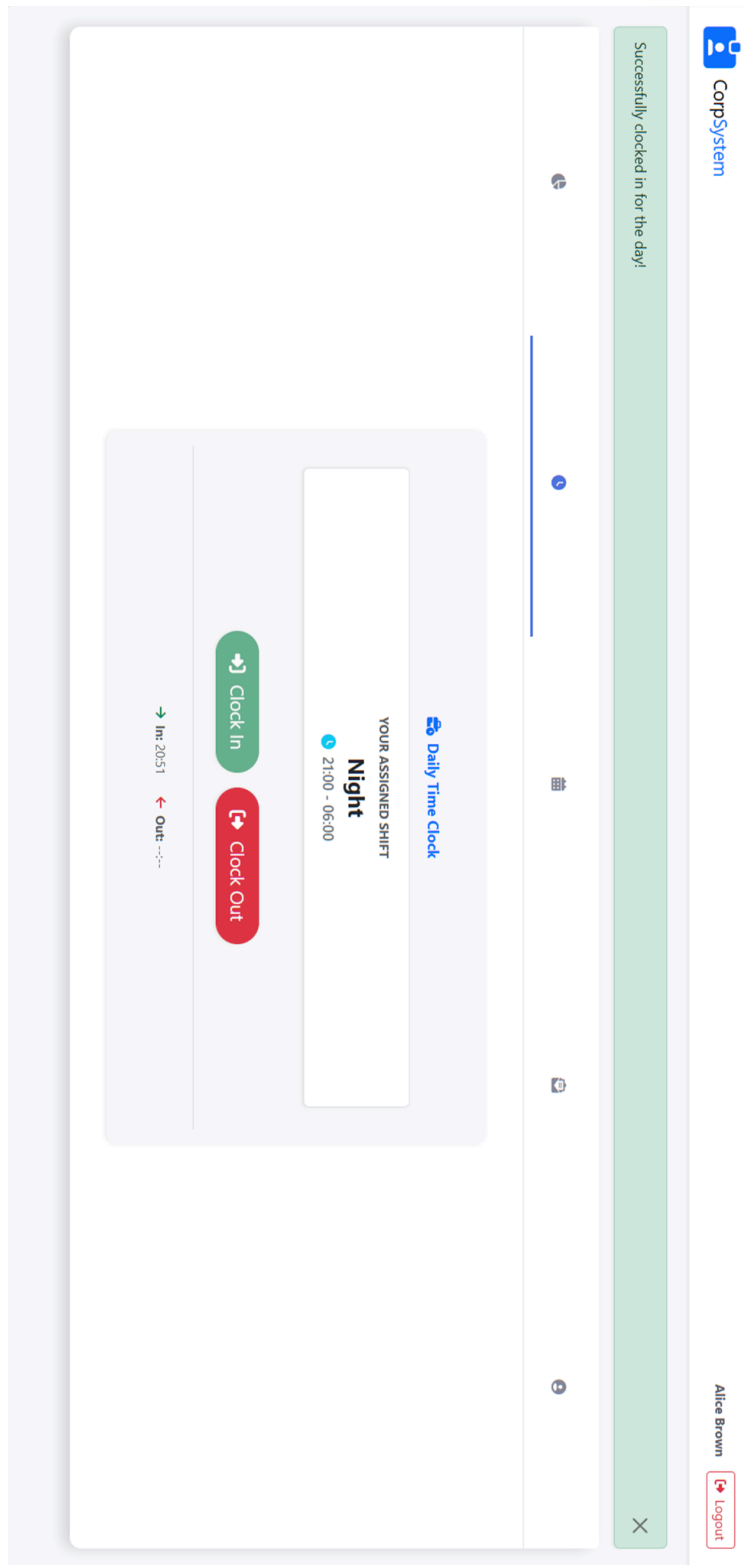


Fig. A-1.12 Employee Module - Clocking

APPENDIX 2: Sample Output Reports

1. Objective

This report analyzes the attendance and leave patterns of Employee_2 to identify trends in full-day and half-day absences, shifts attended, and potential concerns regarding attendance consistency.

2. Attendance Overview

- Total Working Days: 312
- Total Full-Day Present Days: 253
- Total Full-Day Leaves: 28
- Total Half-Day Leaves: 31

3. Monthly Leave Pattern

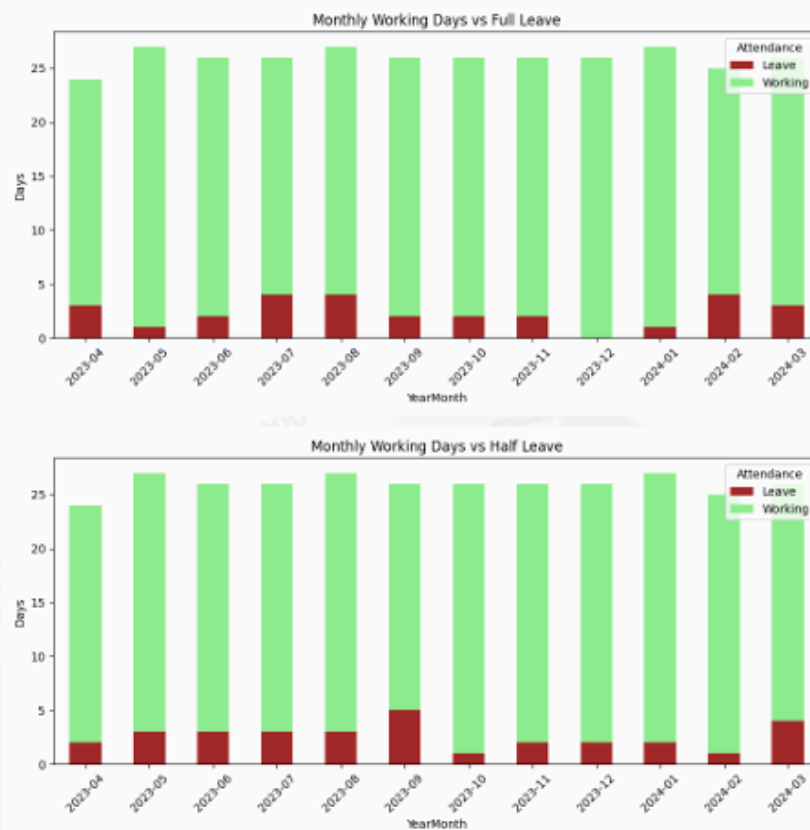


Fig. A-2.1 Analytics Report Page 1

- Month with most Full-Day Leaves: 2024-02
- Month with most Half-Day Leaves: 2023-09

4. Weekday Leave Pattern

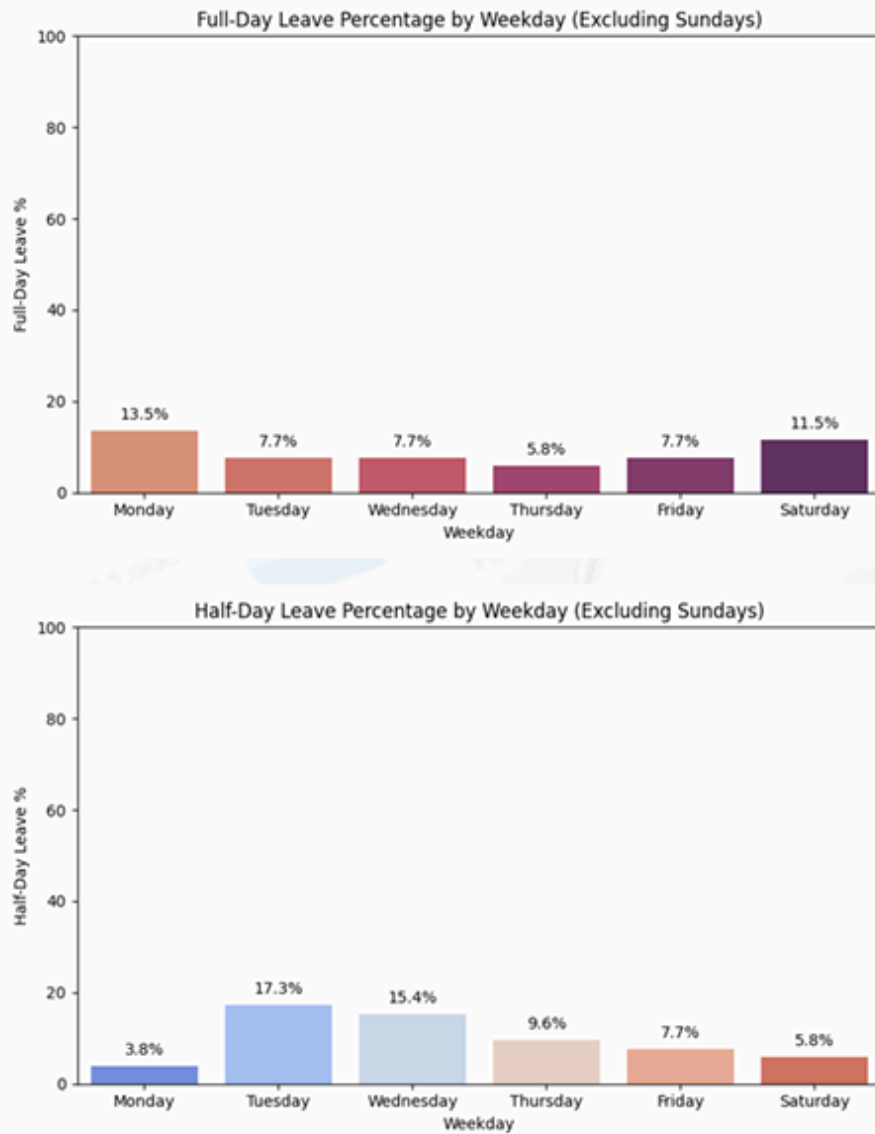


Fig. A-2.2 Analytics Report Page 2

- Weekday with Most Full-Day Leaves: Monday
- Weekday with Most Half-Day Leaves: Tuesday

5. Shift-Based Attendance

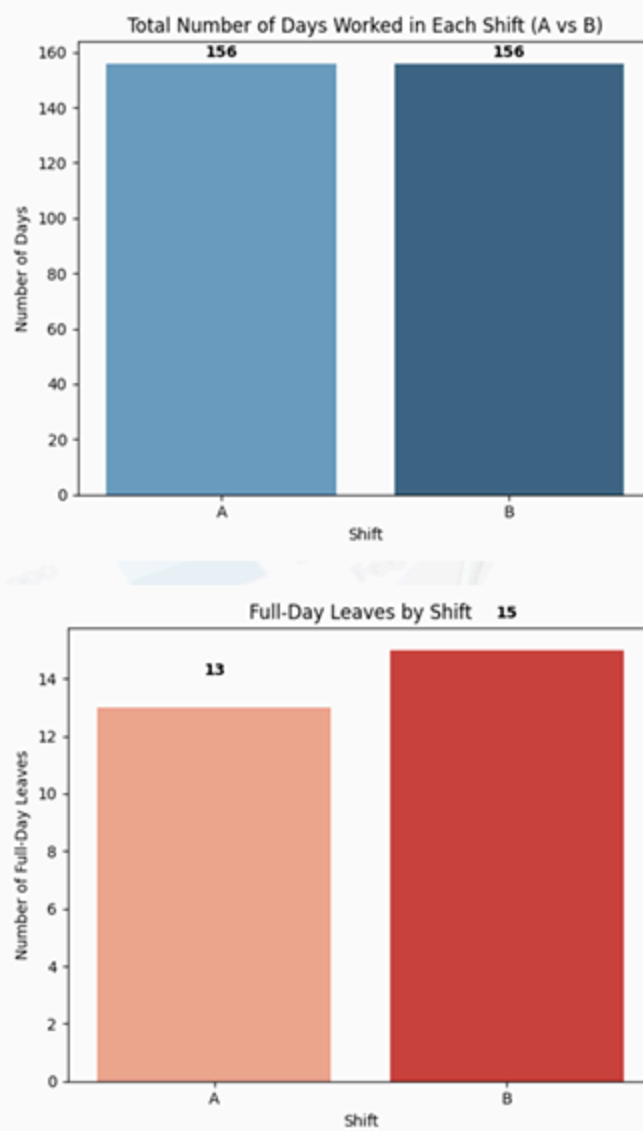


Fig. A-2.3 Analytics Report Page 3

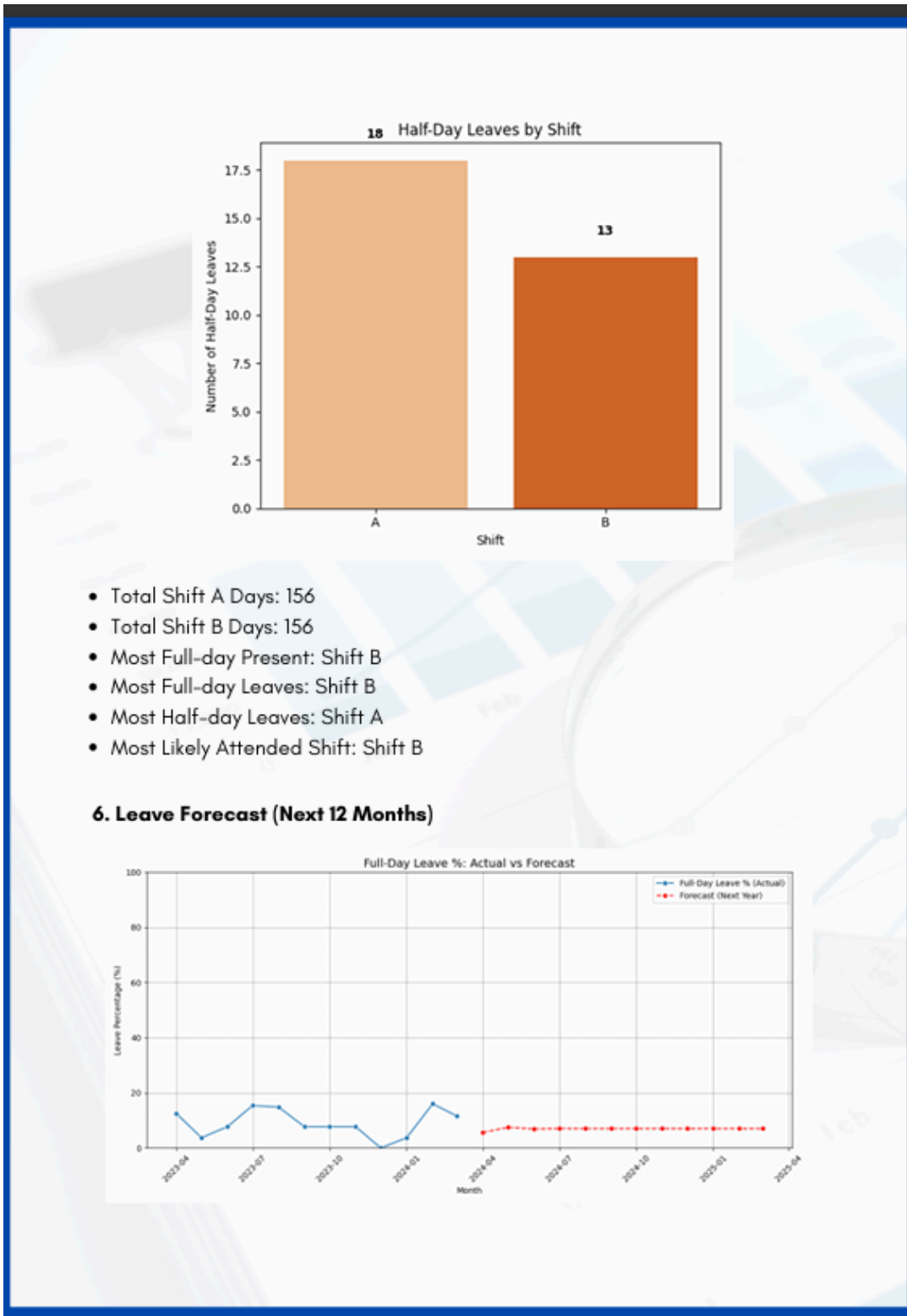


Fig. A-2.4 Analytics Report Page 4



Predicted Full-Day Leaves	Predicted Half-Day leaves
April 2024: 1.62 days	April 2024: 2.46 days
May 2024: 2.04 days	May 2024: 2.59 days
June 2024: 1.91 days	June 2024: 2.58 days
July 2024: 1.95 days	July 2024: 2.58 days
August 2024: 1.94 days	August 2024: 2.58 days
September 2024: 1.94 days	September 2024: 2.58 days
October 2024: 1.94 days	October 2024: 2.58 days
November 2024: 1.94 days	November 2024: 2.58 days
December 2024: 1.94 days	December 2024: 2.58 days
January 2025: 1.94 days	January 2025: 2.58 days
February 2025: 1.94 days	February 2025: 2.58 days
March 2025: 1.94 days	March 2025: 2.58 days

Fig. A-2.5 Analytics Report Page 5

7. Leave Forecast (Next 12 Months)

- The employee tends to take more full-day leaves during **Monday**.
- And more half-day leaves during **Tuesday**.
- **Shift B** shows a higher consistency in attendance.
- Forecast indicates a probable **decrease** in future leaves.

8. Recommendations

Analysis indicates that Employee_2 consistently demonstrates higher attendance during Shift B, suggesting a preference or better compatibility with that shift's timing. Conversely, the employee tends to take more leaves on Mondays and Tuesdays, indicating potential challenges at the start of the workweek.

To mitigate absenteeism and enhance productivity, it is recommended to:

- Implement flexible shift scheduling, prioritizing Shift B allocations for Employee_2 where feasible.
- Schedule high-priority tasks later in the week (from Wednesday onward) to ensure better participation.
- Consider workload balancing or light-duty assignments on Mondays and Tuesdays to ease transitions and improve early-week attendance.

This tailored approach may support better attendance consistency and promote employee well-being without compromising operational needs.

Fig. A-2.6 Analytics Report Page 6

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